Journal of Polymer Research (2026) 33:174
https://doi.org/10.1007/s10965-026-04890-x

ORIGINAL PAPER

Chalcone-filled polyaniline composites for room-temperature thermoelectric applications

Cigdem Yorur Goreci[1] · Volkan Ugraskan[1]

Received: 23 January 2026 / Accepted: 1 April 2026 / Published online: 13 April 2026

Abstract

The effects of the addition of 3,3′-(1,4-phenylene)bis(1-(4-ethoxyphenyl)prop-2-en-1-one) (EtO-BCh), a chalcone derivative, on the thermoelectric (TE) characteristics of polyaniline (PANI) are investigated in this work. XRD patterns revealed that the addition of highly crystalline EtO-BCh leads to interactions that result in structural changes on the PANI backbone, and the distance between molecular layers is reduced. SEM analyses showed the high degree of homogeneity in the composite, which means a massive number of PANI/EtO-BCh interfaces, resulting in scattering low-energy carriers to boost the Seebeck coefficient. FTIR and UV-vis analyses confirmed strong molecular interactions, specifically hydrogen bonding between the carbonyl groups of EtO-BCh and the amine groups of PANI. Hall measurements demonstrated that while the addition of EtO-BCh led to a decrease in carrier concentration from 1.124×10^{20} to 1.2×10^{19} cm^{-3}, it triggered a dramatic seven-fold increase in carrier mobility (increased from 0.98 up to 6.8 cm^2/Vs at 5% EtO-BCh content) due to improved chain alignment. Consequently, the Seebeck coefficient saw a remarkable ten-fold enhancement, rising from 13 to 145 μV/K. Due to this synergistic effect between mobility and the Seebeck coefficient, a maximum power factor of 27.1 μW/mK² was achieved at 5% EtO-BCh content. These results demonstrate that EtO-BCh acts as a highly effective organic modifier for enhancing the TE efficiency of conducting polymers through morphological ordering and energy filtering mechanisms.

Keywords Thermoelectric · Composite · Polyaniline · Chalcone

Introduction

Global energy shortage has become a worldwide issue due to the pace of global industrialization. Concerns about environmental pollution and the depletion of fossil fuels are increasing day by day. Therefore, diversifying energy sources and achieving effective multistage energy usage have become vital solutions for solving global energy and environmental problems. Given the recent surge in energy consumption, the development of new renewable energy sources and the effective use of existing energy technologies are critical. To transform different sources of environmental energy into electrical power, separate energy harvesters are required. The most prevalent energy harvesting methods are piezoelectric, solar, and thermal. However, unlike the other two types of energy harvesting, piezoelectric production requires sufficient mechanical deformation to generate appreciable electrical output, which is difficult to achieve in a car cabin. Furthermore, solar energy collection in the cabin is less effective owing to a lack of light, particularly on overcast, rainy days or at night, which significantly affects efficiency. Furthermore, solar panels require extensive installation space and frequent maintenance, including cleaning, which might be difficult to perform in the cramped confines of a car cabin. In contrast, the engine, exhaust system, and other components generate a huge amount of waste heat during vehicle operation, resulting in a major thermal energy source in the cabin. A variety of waste heat sources, such as manufacturing emissions, residual heat from combustion of fossil fuels, power plants, ships, and oil refineries, may also be utilized. Thermoelectric (TE) power generation allows the direct conversion of heat to electricity without polluting the environment. It provides a unique option for the dependable and sustainable recovery of dissipated waste

✉ Volkan Ugraskan
ugraskan@yildiz.edu.tr

1 Department of Chemistry, Faculty of Arts & Sciences, Yildiz Technical University, 34220 Istanbul, Esenler, Türkiye

heat. Utilizing ambient energy, harvesting industrial waste heat, powering specialized devices, and improving chip-level and personal thermal management are all possible with TE systems [1–7].

TE generators deliver energy conversion with minimal emissions and a high degree of operational reliability over long periods of time. It also has the benefits of no moving parts, silent operation, and environmental friendliness. As a result, these systems could potentially find use in solar cells, space shuttles, the automotive sector, power plants, sensors, monitors, etc. [8–13]. The efficiency of TE materials can be assessed using the dimensionless figure of merit, $ZT = S^2\sigma T\kappa^{-1}$, where S is the Seebeck coefficient, σ is the electrical conductivity, T is the absolute temperature, and κ is the thermal conductivity. Here, $S^2\sigma$ represents the power factor (*PF*). An effective TE material requires high *PF* and low κ [14]. Optimizing the TE properties of current TE materials is one of the main issues, yet there are still additional disadvantages to TE technology. A material's S, σ, and κ are related to one another and rely on the carrier transport mechanism. As a result, it opens up a field of study for simultaneously optimizing these characteristics in a hybrid system of materials [15].

Because of their high ZT values, TE systems currently consist of inorganic compounds like Bi_2Te_3, PbTe, and SiGe, skutterudites, half-Heusler alloys, Zintl phases, clathrates, oxides, metal silicides, liquid-like materials, IV–VI chalcogenides, diamond-like compounds, etc. [16]. However, their large-scale manufacture and uses are restricted by issues including the complexity of the materials, the toxicity of rare elements, and the laborious and time-consuming fabrication process of these TE materials, which contain brittle and heavy hazardous elements [17]. In the case of organic materials, they were undervalued in the early stages of development due to their comparatively low TE performance. However, more recent research has successfully shown that polymer-based TE materials have remarkable TE capabilities comparable to those of traditional inorganic counterparts [18].

The use of organic materials, such as polymers, for thermoelectric applications has gained popularity because of its notable benefits in terms of mechanical flexibility, cost-effective synthesis, and solution processability. In general, polymers have a lower thermal conductivity (usually between 0.1 and 0.5 W/mK) compared to inorganic materials [19]. Conductive polymers, such as poly(3,4-ethylenedioxythiophene) (PEDOT), polyaniline (PANI), polypyrrole (PPy), and their derivatives, enable the regulation of electrical and thermal conductivities by doping. Nonconventional charge and phonon transmission in conjugated polymeric systems offer new chances to improve ZT [20, 21]. PANI, among its relatives, has aroused substantial interest as a TE material because of its excellent environmental stability, low thermal conductivity, and adjustable electrical conductivity. Despite its impressive characteristics, PANI's TE performance is limited by its relatively low Seebeck coefficient. PANI's TE properties are improved by a number of approaches, including post-processing and doping. Besides, the manufacture of nanocomposites has recently acquired popularity in energy applications. The use of suitable additives can also increase the carrier concentration or carrier mobility since the preparation of the physical composites is also an effective strategy [22–25].

Chalcones are common plant-based organic compounds that act as precursors to a wide range of polyphenolic compounds, including flavonoids. Two aromatic rings with a variety of functional groups connected by α- and β-unsaturated carbonyl groups make up the structure of chalcones. They are members of the flavonoid chemical compound family and are naturally occurring derivatives of the original 1,3-diphenyl-2-propen-1-one molecule [26, 27]. Chalcones have drawn a lot of attention due to their simple and reactive chemical structure, potential medicinal properties, and simplicity of production. Chalcones, both synthetic and natural, have a variety of pharmacological properties, including anti-inflammatory, anticancer, antibacterial, antifungal, antimalarial, antidiabetic, and antituberculosis properties. Chalcone derivatives are also being researched extensively for applications in optoelectronic materials, such as non-linear materials, photorefractive polymers, chromophore sensors, and liquid crystal display research [28–30]. The chalcone unit has a special push-pull conjugation action that increases redox activity and luminous properties. The delocalization of electronic charge distribution within these functional molecular systems can increase the mobility of electrons, thereby increasing their density, because of the presence of widely overlapped orbitals where conjugation of electrons may occur efficiently [31]. The symmetrically structured bis-chalcone compound (EtO-BCh) synthesized in this study demonstrates enhanced functionality compared with conventional small-molecule additives commonly employed in PANI-based TE composites. Chalcones, which constitute the structural backbone of this compound, are important biosynthetic intermediates in the formation of naturally occurring flavonoids. Structurally, chalcones are α,β-unsaturated carbonyl compounds that can be readily synthesized under mild reaction conditions and have attracted considerable attention due to their versatile chemical and functional properties. Consequently, chalcone-based structures offer promising potential for applications in functional material systems [32].

Organic additions may be effective TE enhancers, increasing σ and the S while preserving low κ. Culebras et al. reported that bis(trifluoromethylsulfonyl)imide

(BTFMSI)-doped PEDOT: PSS (PEDOT: BTFMSI) had a *PF* of 147 μW/mK² and a measured thermal conductivity of $\kappa=0.19$ W/mK, resulting in a *ZT* of around 0.22 at RT [33]. Deng et al. found that doping PTh derivatives with tetrakis(dimethylamino)ethylene and 4-(2,3-dihydro-1,3-dimethyl-1 H-benzimidazol-2-yl)-N, N-dimethylbenzenamine (N-DMBI) resulted in σ of 144 and 75.4 S/cm, respectively, and *PF* of 75.0 and 98.5 μW/mK² [34]. Mahato et al. found that DMSO-doped PEDOT: PSS exhibited σ of 1230 S/cm [35]. The bis-chalcone compound synthesized in this work acts as an electronically active interfacial modulator, enabling more effective control of the polymer-additive interface and facilitating improved charge carrier mobility within the composite structure.

Recent advances in organic TE materials, specifically PANI, have shifted the focus from simple doping to complex interface engineering and structural ordering to overcome the inverse relationship between σ and the *S* [36]. Furthermore, recent high-performance PANI systems often utilize carbon nanotubes (CNTs) or graphene to create 3D networks. For example, Li et al. achieved a PF of 407 μW/mK² in CNT/PANI films using a sequential dedoping-redoping strategy [37]. This study investigates how to achieve superior *S* with a small-molecule chalcone derivative (EtO-BCh) in contrast to complex carbon-based systems in the literature.

Materials and methods

Terephthalaldehyde (99%), 4-ethoxyacetophenone (98%), potassium hydroxide (powder), aniline (≥99.0%), ammonium persulfate (≥98.0%), hydrochloric acid (37.0%), and ethanol (≥99.9%) were purchased from Sigma Aldrich.

Synthesis of 3,3′-(1,4-phenylene)bis(1-(4-ethoxyphenyl)prop-2-en-1-one) (EtO-BCh)

Bis-chalcone EtO-BCh was synthesized by the Claisen Schmidt condensation of 4-ethoxyacetophenone with terephthalaldehyde following literature procedures [38]. 4-Ethoxyacetophenone (2 equiv.) was dissolved in 10 mL of ethanol, and 2 mL of 40% KOH in water was added dropwise to the reaction mixture at room temperature. Terephthalaldehyde (1 equiv.) was then slowly added to the reaction mixture, and the reaction was stirred at room temperature for 24 h. After the reaction was determined to be complete by TLC, the reaction contents were neutralized with 1 M HCl. The mixture was extracted three times with chloroform, then dried with $MgSO_4$, and the solvent was evaporated. Compound EtO-BCh was then purified by flash column (silica gel) using methylene chloride. The synthesis and chemical structure of EtO-BCh is shown in Fig. 1. Purification resulted pale yellow solid with a yield of 80%. FTIR (ATR): ν (cm^{-1}): 3053 (Ar–CH stretching), 2975 and 2935 (aliphatic CH stretching), 1655 (C=O stretching), 1600, 1577, 1176 (C-O stretching). ^1H NMR (500 MHz, $CDCl_3$): δ (ppm): 8.04 (d, *J*=9.1 Hz, 4 H, Ar–H), 7.79 (d, *J*=15.5 Hz, 2 H, -CH=CH-CO), 7.69 (s, 4 H, Ar–H), 7.60 (d, *J*=15.5 Hz, 2 H, -CO-CH=CH), 6.98 (d, *J*=9.1 Hz, 4 H, Ar–H), 4.13 (q, *J*=7.0 Hz, 4 H, OCH_2), 1.46 (t, *J*=7.0 Hz, 6 H, CH_3).^{13}C NMR (125 MHz, $CDCl_3$): δ (ppm): 187.4 (C=O), 162.0, 141.7, 135.9, 129.9, 129.8, 127.8, 120.4, 113.8, 62.8, 13.6. HRMS (QTOF): *m/z* $[M+H]^+$ calculated for $C_{28}H_{27}O_4$: 427.52; found: 427.1918.

Synthesis of PANI

PANI synthesis was performed in accordance with the literature [39]. First, 20 mL of 1 M HCl was used to dissolve 6.75 mmol of aniline monomer. Dropwise additions of 5 mL of ammonium persulfate solution in 1 M HCl were made to the polymerization medium. At RT, the polymerization process lasted 14 h. To get rid of any unreacted chemicals, PANI was repeatedly washed with ethanol and distilled water after polymerization.

Preparation of the composites

PANI and chalcone were mechanically mixed to create the composites. First, the composite powders with various amounts of chalcone (1%, 3%, and 5%) were mixed together. The components were mechanically mixed in a mortar under ambient conditions for 15 min. The powders were ground into pellets with a diameter of 1 cm and a thickness of 1 mm using a mechanical press at a pressure of 100 MPa at RT in order to conduct TE experiments on pure PANI and composites.

Characterization

A Panalytical Empyrean X-ray diffractometer (anode material: Cu; intended wavelength type: K-α1 = 1.54 Å; generator settings: 40 mA, 45 kV; step size [°2θ]=0.0130) was used for the X-ray diffraction (XRD) investigations. The Zeiss EVO LS 10 scanning electron microscope was used to perform SEM-EDS studies. FTIR-ATR (Fourier transform infrared spectroscopy-attenuated total reflectance) analyses were conducted using a NICOLET IS10 Thermo FTIR instrument. A Shimadzu UV2600 UV-vis spectrophotometer with a wavelength range of 300–1000 nm was used to measure the optical properties of the materials. A Hall system (Ecopia HMS-3300) was used to assess mobility characteristics, charge carrier concentration, and

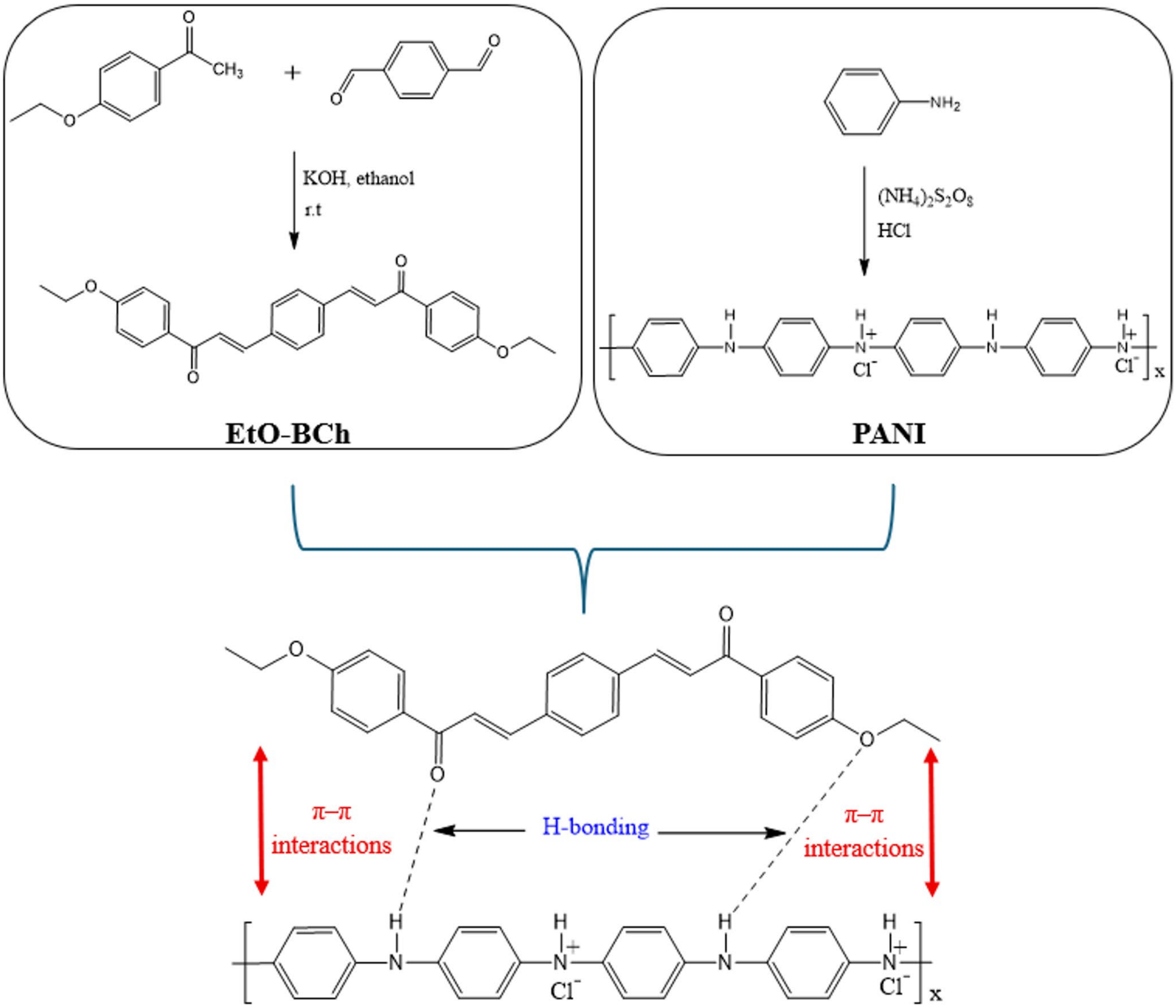


Fig. 1 Schematic representation of synthesis of PANI and EtO-BCh

electrical conductivity. The Entek Electronic Seebeck Coefficient Measuring System was used to calculate Seebeck coefficients. ^{1}H and ^{13}C NMR spectra were recorded on a Bruker Avance spectrometer operating at 500 and 125 MHz, respectively, using $CDCl_3$ as the solvent and tetramethylsilane (TMS) as the internal reference. High-resolution mass spectrometry (HRMS) analyses were conducted on an Agilent 6530 LC-QTOF system.

Results and Discussion

Figure 2 displays the XRD patterns of the EtO-BCh and PANI/ EtO-BCh composite that contains 5% EtO-BCh. The sharpness of the peaks for EtO-BCh suggests a high level of crystalline nature. Additionally, the sharp peaks between 2θ = 2–10° represent long-range lamellar (layered) ordering [40]. The typical peaks at 2θ = 15.3°, 20.2°, and 25.7° in the pure PANI pattern are indicative of the semi-crystalline structure of the emeraldine salt (ES) form of PANI [41]. The presence of EtO-BCh in the composite structure was verified by the peaks observed in the composite pattern. Additionally, there was a minor shift in the diffraction peaks. The peaks of pure PANI at 2θ = 15.3°, 20.2°, and 25.7° shifted to 15.33°, 21.3°, and 25.86°, increasing by 0.03°, 1.1°, and 0.16°, respectively. These values are 2–8 times larger than the step size, therefore it is not a random artifact. According to Bragg's law ($n\lambda \ = 2dsin\theta$), it was observed that the interplanar spacing (*d*) values decreased and the PANI lattice contracted as a result of shifts to higher 2θ values. This situation indicates that interactions leading to structural changes on the PANI backbone occur with the addition

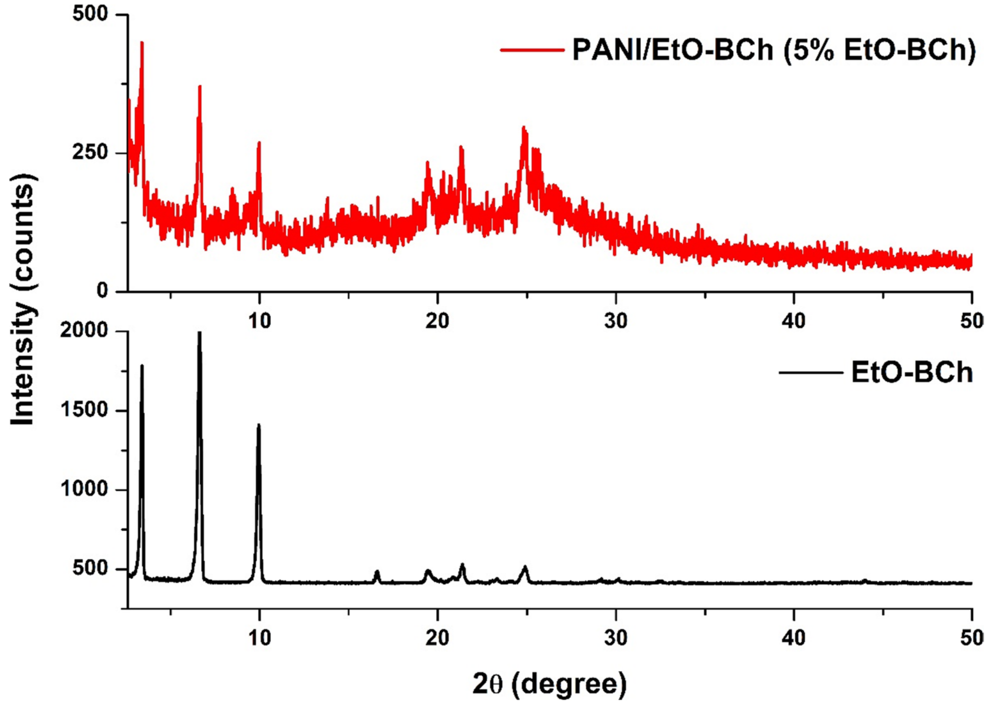


Fig. 2 XRD patterns of the EtO-BCh and PANI/ EtO-BCh composite containing 5% EtO-BCh

of EtO-BCh, and the distance between molecular layers is decreasing [42].

The crystallite sizes of EtO-BCh and PANI/ EtO-BCh composite were calculated using the Debye-Scherrer equation:

$$D = \frac{K\lambda}{\beta\, cos\theta} \tag{1}$$

where the average nanocrystal diameter (nm), corrected band broadening (FWHM), crystallite size constant (0.9), X-ray wavelength, and diffraction angle are represented by the parameters D, β, K, λ, and θ, respectively [43]. According to data, the crystallite sizes of EtO-BCh and the composite were found to be 51.6 nm and 48.81 nm. TE performance is significantly impacted by a material's grain size. Achieving a nanocrystallite size improves TE performance by increasing phonon scattering at interfaces [44]. Therefore, it was anticipated that the nanoscale structure of EtO-BCh grains would enhance PANI's TE characteristics.

SEM micrographs of the EtO-BCh and PANI/EtO-BCh composite containing 5% EtO-BCh are illustrated in Fig. 3. Analyses revealed that the EtO-BCh is composed of flake-like particles with rounded edges and in different sizes (Fig. 3a) [45]. The micrograph of the composite confirms that the EtO-BCh is homogeneously dispersed within the PANI matrix. The uniform distribution of the granular PANI suggests a high degree of distributional homogeneity. There is no obvious phase separation or large "clumping" of the

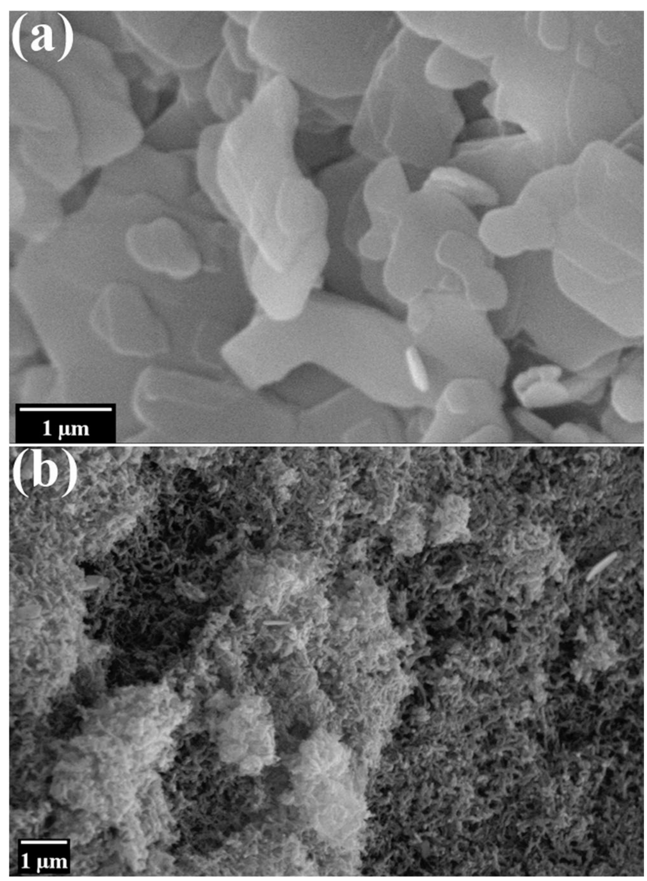


Fig. 3 SEM micrographs of the EtO-BCh and PANI/EtO-BCh composite containing 5% EtO-BCh

5% EtO-BCh content, which indicates that the mixing process effectively integrated the components at the microscale (Fig. 3b).

FTIR spectra of the pure PANI, EtO-BCh, and PANI/EtO-BCh composite containing 5% EtO-BCh are shown in Fig. 4. In the FTIR spectrum obtained to identify the

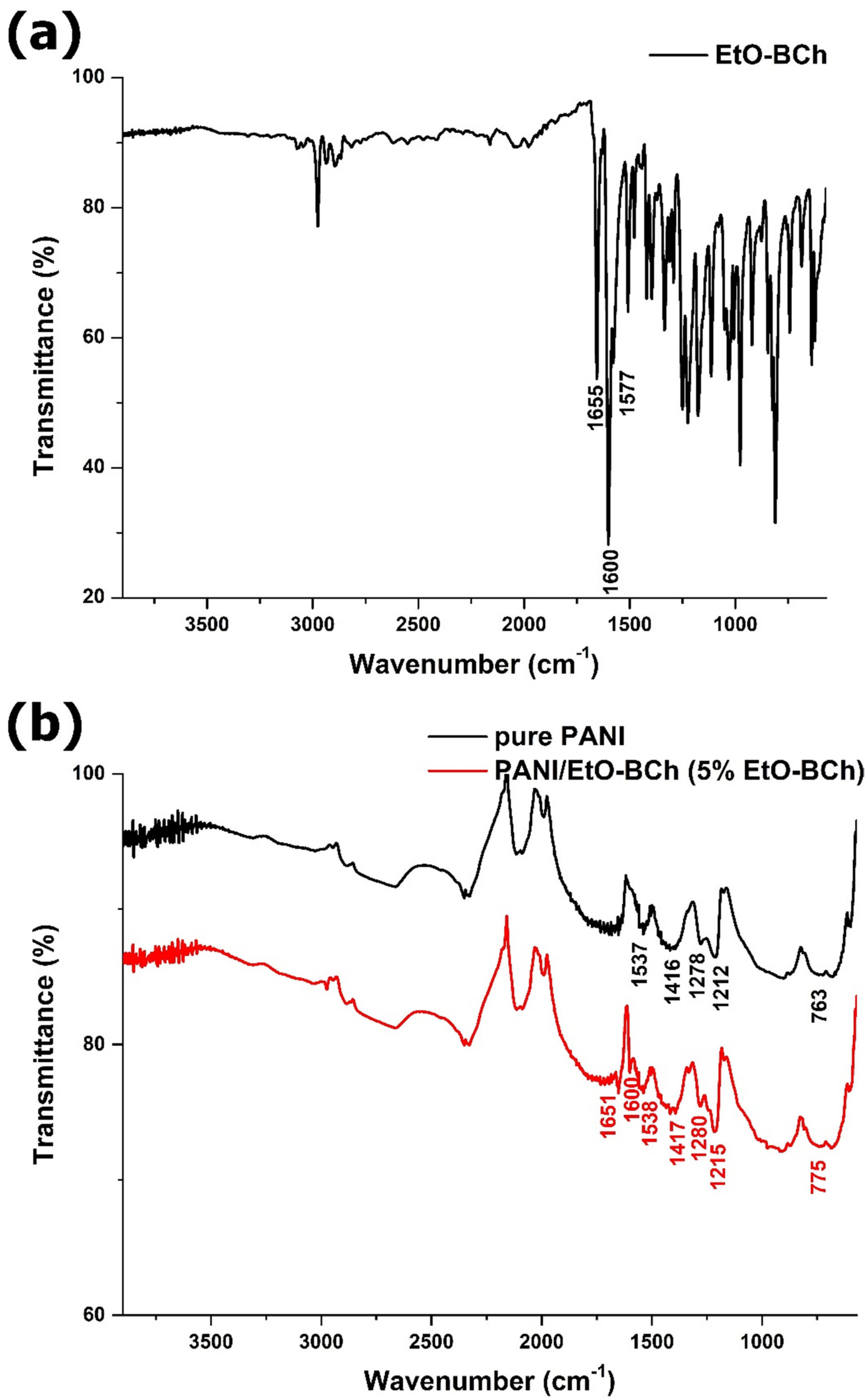


Fig. 4 FTIR spectra of the pure PANI, EtO-BCh and PANI/EtO-BCh composite containing 5% EtO-BCh

prominent functional groups in the structural analysis of the chalcone compound EtO-BCh, the stretching vibration characteristic of the carbonyl (-C=O) group, typical for chalcone compounds, was observed at 1655 cm^{-1}. The stretching vibrations of the aliphatic -C=C- stretching vibrations adjacent to the carbonyl group were observed at 1600–1577 cm^{-1} (Fig. 4a) [46]. The peaks found at 1537 and 1416 cm^{-1} in the pure PANI spectrum demonstrated the presence of characteristic peaks of PANI, specifically the stretching of quinoid and benzenoid rings. The C-N/C=N stretching modes were linked to the peak at 1278 cm^{-1}, and the peak of the protonated C-N group (C-N^+-H) was observed at 1212 cm^{-1}. An absorption band formed by the C–H out-of-plane deformation of a 1,2-disubstituted aromatic ring was seen at 763 cm^{-1} (Fig. 4b) [47–49].

The composite spectrum shows a new peak at 1651 cm^{-1}, which corresponds to the C=O stretching vibration. This demonstrates the successful incorporation of the oxygen-containing EtO-BCh filler, which serves as a proton acceptor for hydrogen bonding. Another notable feature in this spectrum is the blue shift of multiple peaks in the composite compared to those in pure PANI, which indicates bond shrinkage and strengthening. For example, peaks at 1212 cm^{-1} and 1278 cm^{-1} shifted up to 1215 cm^{-1} and 1280 cm^{-1}, respectively. According to the literature, these blue shifts suggest bond shrinkage and strengthening, which are frequently connected with "improper" H-bonding, where electron density redistribution stabilizes the polymer backbone. H-bonding possibly formed between the amino (-NH-) groups of the PANI backbone and oxygen-containing functional groups (ethoxy or carbonyl). The composite spectrum exhibits reduced transmittance (increased absorption) for multiple peaks, especially in the 1200–1600 cm^{-1} range. This rise in peak intensity is a common "fingerprint" that confirms the formation of new intermolecular contacts, such as H-bonds [50–53]. The formation of H-bonds and the increasing compactness of the structure observed in FTIR studies are consistent with XRD data.

UV-vis spectra of pure PANI and PANI/EtO-BCh composites are shown in Fig. 5. The emeraldine salt form of PANI has a local absorption peak around 340 nm, which corresponds to the π-π* transition of the benzene ring. The bands in the visible range at 435 and 849 nm are associated with the presence of polaron states and attributed to π−polaron and polaron−π* transitions [54]. The composites' spectra revealed that the peaks associated with π-π* transitions shifted toward higher wavelength values, a phenomenon known as bathochromic shift. This shift is connected to changes in conjugation lengths [55]. This could be owing to interactions between PANI chains and EtO-BCh, which facilitate charge transfer via H-bonding [56]. These shifts also correlate to easy delocalization of π-electrons, which supports a π-cloud-electronic band interaction between PANI and EtO-BCh particles [57].

The σ is proportional to carrier concentration (n), electronic charge (e), and carrier mobility (μ) ($\sigma = n\, e\, \mu$) [42]. In general, high carrier concentration and/or mobility enable the attainment of high electrical conductivity [58]. However, a large n value causes a drop in the S. Figure 6 shows the Hall parameters for pure PANI and PANI- EtO-BCh composites. Data shows that

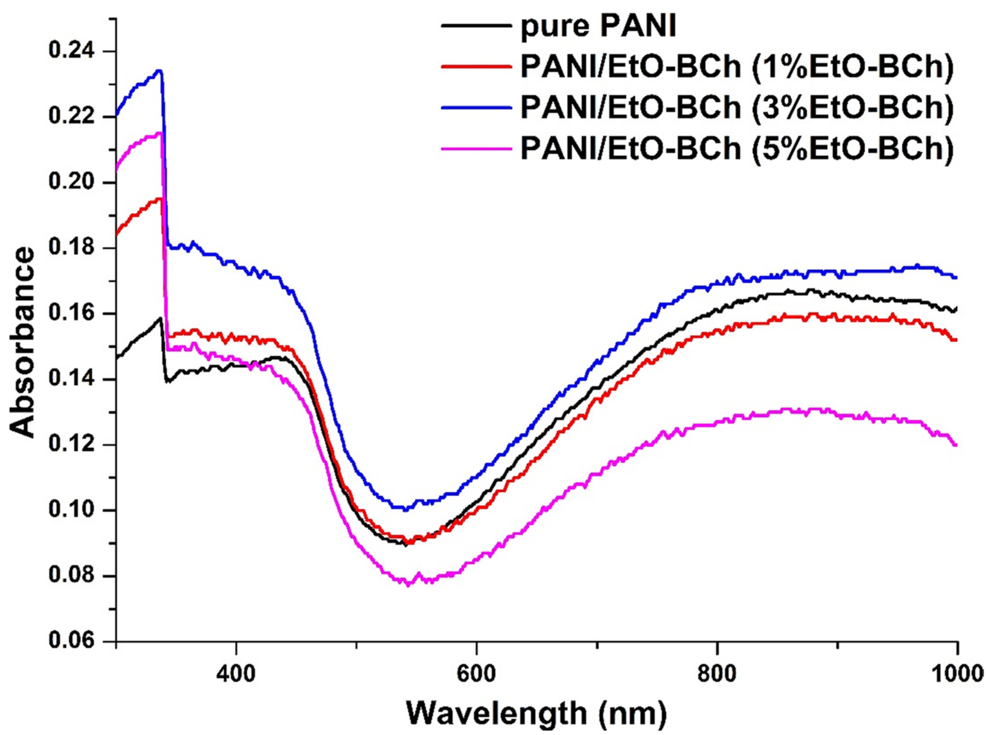
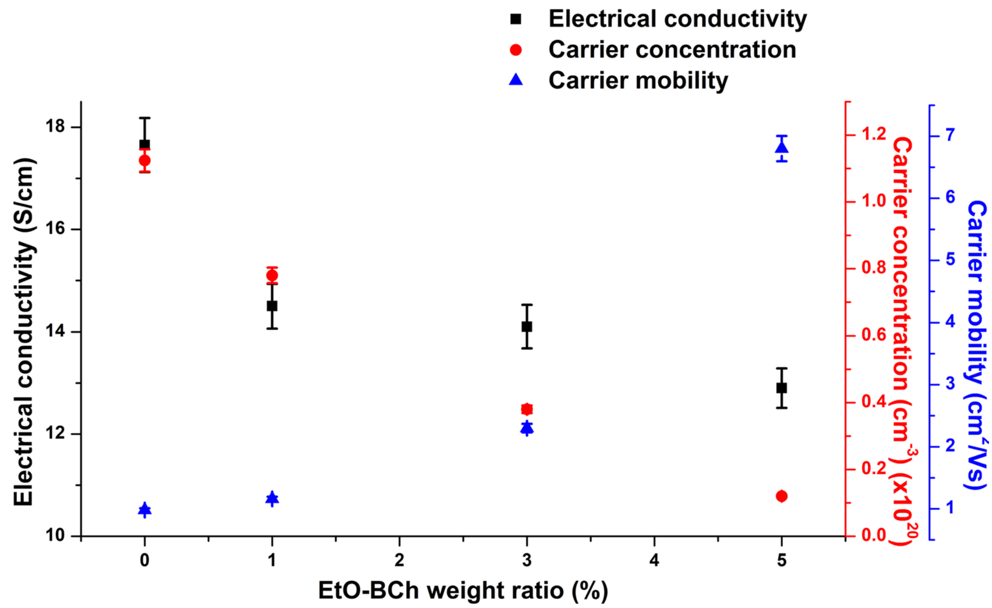


Fig. 5 UV-vis spectra of the pure PANI, EtO-BCh and PANI/ EtO-BCh composites

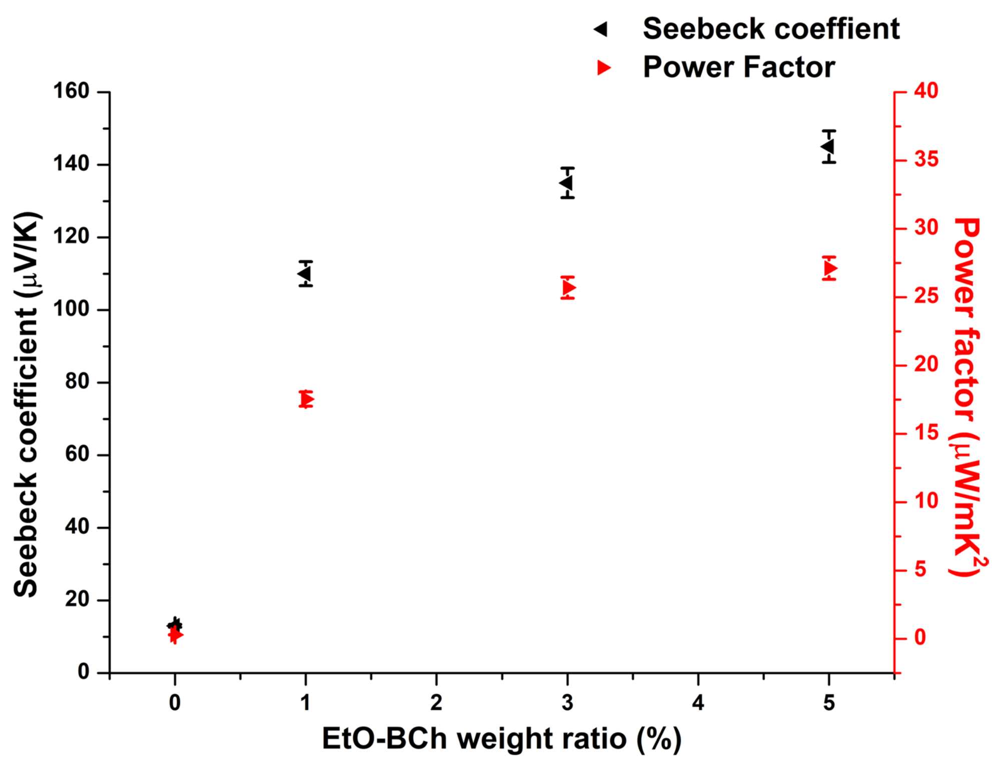
Fig. 6 σ, n, and μ values of the pure PANI and PANI/EtO-BCh composites



increasing the amount of EtO-BCh leads to an increase in μ and a decrease in n. As mentioned in XRD and FTIR observations, lattice contraction and stronger H-bonding result in a more ordered and compact structure, allowing charge carriers to travel more easily and faster down the chains. The addition of EtO-BCh facilitates the adsorption of PANI layers on their surfaces, promoting charge transfers and broadening the polymeric molecular conformation, packing the polymeric chains in a high order, and resulting in an increase in μ [59]. Additionally, the σ of pure PANI fell from 17.6 S/cm to 12.9 S/cm as the EtO-BCh content increased. The decrease is mostly the effect of the substantial reduction of n. This might be the result of the de-doping effect of EtO-BCh. The oxygen-containing groups in the EtO-BCh are anticipated to interact with the protonated nitrogen sites of the PANI. This interaction "neutralizes" or traps certain charge carriers, hence limiting the quantity of accessible holes/electrons for conduction.

S analyses demonstrated that all composites exhibited positive S values, which are indicative of *p-type* conductors (Fig. 7). Composite samples with 5% EtO-BCh had the largest S of 145

Fig. 7 S and PF values of the pure PANI and PANI/EtO-BCh composites



μV/K, indicating a direct correlation with EtO-BCh content (Table 1). The rise in *S* values, as previously stated, is attributable to improved mobility in the composites. The EtO-BCh considerably enhanced carrier mobility by bridging carrier transport between PANI chains. Furthermore, as observed in SEM micrographs, the interface is highly homogeneous, with PANI coating EtO-BCh in a granular manner. These interfaces may act as a barrier. These barriers may serve as "filters" that scatter low-energy charge carriers while enabling high-energy (hot) carriers to flow through. Filtering out low-energy carriers considerably improves the *S*, which measures the average energy delivered per charge carrier. As a result, S was significantly improved [60]. *PF* values were calculated using σ and *S* data ($PF = S^2\sigma$). The addition of 5% EtO-BCh increased the *PF* value of pure PANI by around 90-fold, from 0.3 to 27.1 μW/mK². The κ of the pure PANI was reported as 0.14 W/mK [61]. According to the literature data, κ for the organic molecules lies between 0.12 and 0.35 W/mK [62]. A basic model for thermal conductivity of materials that avoids microscopic details is based on the Kapitza resistance and the effective medium approximation. In an effective medium method, a material's overall κ is calculated as the sum of its intra- and inter-grain contributions. The intergrain component results from the interfacial resistance, also known as the Kapitza resistance, to heat transmission. In the presence of a thermal gradient, the Kapitza resistance causes a temperature discontinuity across the contact, which Kapitza initially identified. The overall impact of Kapitza resistance is to lower heat conductivity, which is more prominent at smaller grain sizes [63]. As discussed in *S* data, high Kapitza resistance interfaces can act as energy filters, blocking low-energy carriers, which increases the average energy per carrier and thus boosts the *S*. As a result, it is expected that the composites will possess lower or similar thermal conductivities compared to those of pure PANI.

The transport properties of the samples are summarized in Table 1. To ensure the reproducibility of the data and account for variations in material processing, all measurements were performed on three independently prepared pellets for each composition. The resulting error bars represent the standard deviation across these replicates, demonstrating high consistency in both the synthesis and pelletization processes.

Table 1 TE parameters of the pure PANI and PANI/NiS composites at RT

Sample	n (cm^{-3}) (x10^{20})	μ (cm^2/Vs)	σ (S/cm)	S (μV/K)	PF (μW/mK^2)
Pure PANI	1.124±0.03	0.98±0.03	17.6±0.5	13±0.4	0.3±0.01
PANI/EtO-BCh (1.0%)	0.78±0.02	1.16±0.04	14.5±0.4	110±3.3	17.5±0.5
PANI/EtO-BCh (3.0%)	0.38±0.01	2.3±0.07	14.1±0.5	135±4.1	25.7±0.8
PANI/EtO-BCh (5.0%)	0.12±0.004	6.8±0.2	12.9±0.4	145±4.35	27.1±0.8

Conclusion

This study examines the TE performance of EtO-BCh-filled PANI composites. EtO-BCh was synthesized for this purpose, and composites of 1%, 3%, and 5% EtO-BCh were produced. The XRD results indicated slight modifications for the composites, demonstrating that EtO-BCh was effective on the crystalline characteristics of PANI. The uniform distribution of EtO-BCh in the composite structure was revealed by SEM studies. FTIR and UV-vis investigations demonstrated the formation of H-bonds between EtO-BCh and PANI. Hall measurements indicated that increasing the amount of EtO-BCh leads to an increase in μ and a reduction in n. With the addition of EtO-BCh, the σ of PANI reduced from 17.6 S/cm to 12.9 S/cm, since the decrease in n was larger than the rise in μ. Furthermore, the *S* of pure PANI rose from 13 to 145 μV/K, which is precisely proportional to the rise in μ. The *PF* calculations showed that the pure PANI rose from 0.3 to 27.1 μW/mK². This study confirms the high thermoelectric potential of PANI/EtO-BCh composites and, in light of these findings, identifies future research and application areas such as wearable electronics, self-powered sensors, sensitive temperature sensors. To improve the final efficiency and industrial suitability of the study, effect of the EtO-BCh structure on phonon scattering, and the figure of metit (ZT) of the material will be calculated precisely. The obtained optimum composite will be printed onto flexible substrates to transform it into multilayer organic thermoelectric generator (OTEG) prototypes. The electrical performance of the material under different humidity and temperature conditions will be systematically examined (aging tests).

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10965-026-04890-x.

Authors' Contribution Cigdem Yorur Goreci: Formal analysis, Investigation, Methodology, Writing – original draft. Volkan Ugraskan: Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Funding The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Sun Q, Du C, Chen G (2025) Thermoelectric materials and applications in buildings. Prog Mater Sci. https://doi.org/10.1016/j.pmatsci.2024.101402
2. Choo S, Lee J, Şişik B, Jung SJ, Kim K, Yang SE, Jo S, Nam C, Ahn S, Lee HS, Chae HG, Kim SK, LeBlanc S, Son J (2024) Geometric design of Cu_2Se-based thermoelectric materials for enhancing power generation. Nat Energy. https://doi.org/10.1038/s41560-024-01589-5
3. Shi XL, Li NH, Li M, Chen ZG (2025) Toward efficient thermoelectric materials and devices: advances, challenges, and opportunities. Chem Rev. https://doi.org/10.1021/acs.chemrev.5c00060
4. Wu C, Shi XL, Wang L, Lyu W, Yuan P, Cheng L, Chen ZG, Yao X (2024) Defect engineering advances thermoelectric materials. ACS Nano. https://doi.org/10.1021/acsnano.4c11732
5. Hu X, Quan B, Shi Z, Zhao X, Lu G, Ding Y, Lai J, Qu J, Peng Y, Lu X (2026) Bioinspired Hierarchical Radiative-Phase Change Hybrid Cooling Composite with Record-Breaking Cooling Power. Adv Mater. https://doi.org/10.1002/adma.202510988
6. Zhan Z, Lin L, Zhang J, Li Z, Zhang Q, Wang Q (2025) Effect of In Situ Polymerization on the Thermal Stability and Enthalpy Properties of Wood-Based Composites Based on Graphene Oxide–Modified Phase Change Microcapsule. Polym Compos. https://doi.org/10.1002/pc.30133
7. Wang H, Niu A, Waktole DA, Jia B, Zuo Z, Wang W (2025) A self-powered wireless temperature sensing system using flexible thermoelectric generators under simulated thermal condition. Measurement. https://doi.org/10.1016/j.measurement.2025.117637
8. Singh R, Dogra S, Dixit S, Vatin NI, Bhardwaj R, Sundramoorthy AK, Perera HCS, Patole SP, Mishra RK, Arya S (2024) Advancements in thermoelectric materials for efficient waste heat recovery and renewable energy generation. Hybrid Adv. https://doi.org/10.1016/j.hybadv.2024.100176
9. Moshwan R, Shi XL, Liu WD, Liu J, Chen ZG (2024) Entropy engineering: An innovative strategy for designing high-performance thermoelectric materials and devices. Nano Today. https://doi.org/10.1016/j.nantod.2024.102475
10. Xiong Q, Han G, Wang G, Lu X, Zhou X (2024) The doping strategies for modulation of transport properties in thermoelectric materials. Adv Funct Mater. https://doi.org/10.1002/adfm.202411304
11. Miao Z, Meng X, Li X, Liang B (2025) Influence weights of key parameters and optimization strategies for non-contact thermoelectric generator performance enhancement. Energy. https://doi.org/10.1016/j.energy.2025.139120
12. Liu S, Fan F, Wang A, Xu W, Zhao L, Chen R (2026) 3D simulation and performance analysis of the non-isothermal charge/discharge processes in thermo-electrochemical cycle. J Power Sources. https://doi.org/10.1016/j.jpowsour.2025.238695
13. Cai Y, Jin P, Wang X, Chen C, Fu H, Huang J, Deng F (2025) Nature-inspired wearable thermoelectric generator for body heat harvesting. Energy. https://doi.org/10.1016/j.energy.2025.138899
14. Cao X, Shi XL, Zou J, Chen ZG (2021) Advances in conducting polymer-based thermoelectric materials and devices. Microstructures. https://doi.org/10.20517/microstructures.2021.06
15. Bhagade S, Debnath A, Das D, Saha B (2022) Thermoelectric composite material of $CuBO_2$ incorporated PANI powders with enhanced Seebeck coefficient. J Polym Res. https://doi.org/10.1007/s10965-022-03306-w
16. Deng T, Qiu P, Yin T, Li Z, Yang J, Wei T, Shi X (2024) High-throughput strategies in the discovery of thermoelectric materials. Adv Mater. https://doi.org/10.1002/adma.202311278
17. Hossain S, Yamamoto Y, Baba S, Sakai S, Kishi N (2024) Role of anionic surfactant addition in improving thermoelectric properties of PEDOT:PSS free-standing films. J Polym Res. https://doi.org/10.1007/s10965-024-04078-1
18. Abbasi MS, Sultana R, Ahmed I, Adnan M, Shah UA, Irshad MS, Vu HN, Do LT, Vu HHT, Pham TD, Nang HX, Dao VD (2024) Contemporary advances in organic thermoelectric materials: Fundamentals, properties, optimization strategies, and applications. Renew Sustain Energy Rev. https://doi.org/10.1016/j.rser.2024.114579
19. Li Y, Bao C, Dong L, Wang Y, Wu Z, Xie H (2025) Enhanced thermoelectric properties of PEDOT: PSS self-supporting films prepared by ethanol gel- film formation method with acidic and alkaline post-treatments. J Polym Res. https://doi.org/10.1007/s10965-025-04665-w
20. Armida SA, Ebrahimibagha D, Ray M, Datta S (2024) Assessing thermoelectric performance of quasi 0D carbon and polyaniline nanocomposites using machine learning. Adv Compos Mater. https://doi.org/10.1080/09243046.2023.2262875
21. Song E, Liu P, Lv Y, Wang E, Guo CY (2024) Conductive Polymer-Based Thermoelectric Composites: Preparation, Properties, and Applications. J Compos Sci. https://doi.org/10.3390/jcs8080308
22. Singh M, Gautam AK, Faraz M, Khare N (2024) Polyaniline/graphitic carbon nitride/reduced graphene oxide ternary nanocomposite film for flexible thermoelectric application. Nanotechnology. https://doi.org/10.1088/1361-6528/ad7b3e
23. Zhang L, Shi XL, Yang YL, Chen ZG (2021) Flexible thermoelectric materials and devices: From materials to applications. Mater Today. https://doi.org/10.1016/j.mattod.2021.02.016
24. Wu Y, Ganguly S, Tang XS (2024) 3D bioprinting of anisotropic filler-reinforced polymer nanocomposites: Synthesis, assembly, and multifunctional applications. Int J Bioprinting. https://doi.org/10.36922/ijb.1637
25. Lu JY, Bu ZQ, Lei YQ, Wang D, He B, Wang J, Huang WT (2024) Facile microwave-assisted synthesis of Sb_2O_3-CuO nanocomposites for catalytic degradation of p-nitrophenol. J Mol Liq. https://doi.org/10.1016/j.molliq.2024.125503
26. Espinoza-Hicks JC, Nápoles-Duarte JM, Nevárez-Moorillón GV, Camacho-Dávila A, Rodríguez-Valdez LM (2016) Synthesis, electronic, and spectral properties of novel geranylated chalcone derivatives: a theoretical and experimental study. J Mol Model. https://doi.org/10.1007/s00894-016-3114-x
27. Hernández-Ortiz OJ, Cerón-Castelán JE, Muñoz-Pérez FM, Salazar-Pereda V, Ortega-Mendoza JG, Veloz-Rodríguez MA, Lobo-Guerrero A, Espinosa-Roa A, Rodríguez-Rivera MA, Vázquez-García RA (2020) Synthesis, optical, electrochemical, and magnetic properties of new ferrocenyl chalcone semiconductors for optoelectronic applications. J Mater Sci: Mater Electron. https://doi.org/10.1007/s10854-020-02882-1
28. Basappa VC, Ramaiah S, Penubolu S, Kariyappa AK (2021) Recent developments on the synthetic and biological applications of chalcones—A review. Biointerface Res Appl Chem. https://doi.org/10.33263/BRIAC121.180195
29. Nematollahi MH, Mehrabani M, Hozhabri Y, Mirtajaddini M, Iravani S (2023) Antiviral and antimicrobial applications of chalcones and their derivatives: From nature to greener synthesis. Heliyon. https://doi.org/10.1016/j.heliyon.2023.e20428
30. Singh G, Satija P, Lin FS, Kaur J, Ho KC (2022) New energy harvesting using conjugated chalconyl-organosiloxyl framework. Mater Chem Phys. https://doi.org/10.1016/j.matchemphys.2022.125751
31. Khairul WM, Hashim F, Rahamathullah R, Mohammed M, Razali SA, Johari SATT, Azizan S (2024) Exploring ethynyl-based chalcones as green semiconductor materials for optical

limiting interests. Spectrochim Acta-A: Mol Biomol Spectrosc. https://doi.org/10.1016/j.saa.2023.123776
32. Chen M, Xie D, Zhou H, Zong P (2025) PANI-Based Thermoelectric Materials. Organics. https://doi.org/10.3390/org6030033
33. Culebras M, Gómez CM, Cantarero A (2014) Enhanced thermoelectric performance of PEDOT with different counter-ions optimized by chemical reduction. J Mater Chem A. https://doi.org/10.1039/C4TA01012D
34. Deng S, Liu J, Meng B, Liu J, Wang L (2023) A Highly Conductive n-Type Polythiophene Derivative: Effect of Molecular Weight on n-Doping Behavior and Thermoelectric Performance. ACS Appl Mater Interfaces. https://doi.org/10.1021/acsami.3c10601
35. Mahato S, Puigdollers J, Voz C, Mukhopadhyay M, Mukherjee M, Hazra S (2020) Near 5% DMSO is the best: A structural investigation of PEDOT: PSS thin films with strong emphasis on surface and interface for hybrid solar cell. Appl Surf Sci. https://doi.org/10.1016/j.apsusc.2019.143967
36. Liu S, Li H, Li P, Liu Y, He C (2021) Recent Advances in Polyaniline-Based Thermoelectric Composites. CCS Chem. https://doi.org/10.31635/ccschem.021.202101066
37. Li H, Liu Y, Li P, Liu S, Du F, He C (2021) Enhanced Thermoelectric Performance of Carbon Nanotubes/Polyaniline Composites by Multiple Interface Engineering. ACS Appl Mater Interfaces. https://doi.org/10.1021/acsami.0c20931
38. Chen S, Qin C, Jin M, Pan H, Wan D (2021) Novel chalcone derivatives with large conjugation structures as photosensitizers for versatile photopolymerization. J Polym Sci. https://doi.org/10.1002/pol.20210024
39. Tang SJ, Wang AT, Lin SY, Huang KY, Yang CC, Yeh JM, Chiu KC (2011) Polymerization of aniline under various concentrations of APS and HCl. Polym J. https://doi.org/10.1038/pj.2011.43
40. Vanderstraeten H, Neerinck D, Temst K, Bruynseraede Y, Fullerton EE, Schuller IK (1991) Low-angle X-ray diffraction of multilayered structures. J Appl Crystallogr. https://doi.org/10.1107/S0021889891004156
41. Mitra M, Kulsi C, Chatterjee K, Kargupta K, Ganguly S, Banerjee D, Goswami S (2015) Reduced graphene oxide-polyaniline composites—synthesis, characterization and optimization for thermoelectric applications. RSC Adv. https://doi.org/10.1039/C5RA01794G
42. Alzoubi F, Al-Gharram M, AlZoubi T, Al-Khateeb H, Al-Qadi M, Abu Noqta O, Makhadmeh G, Mouhtady O, Al-Hmoud M, Mandumpal J (2025) Tailoring CuO/Polyaniline Nanocomposites for Optoelectronic Applications: Synthesis, Characterization, and Performance Analysis. Polymers. https://doi.org/10.3390/polym17101423
43. Svoboda L, Praus P, Lima MJ, Sampaio MJ, Matýsek D, Ritz M, Dvorský R, Faria JL, Silva CG (2018) Graphitic carbon nitride nanosheets as highly efficient photocatalysts for phenol degradation under high-power visible LED irradiation. Mater Res Bull. https://doi.org/10.1016/j.materresbull.2017.12.049
44. Ngo TNM, Palstra TTM, Blake GR (2016) Crystallite size dependence of thermoelectric performance of $CuCrO_2$. RSC Adv. https://doi.org/10.1039/C6RA08035A
45. Jarag KJ, Pinjari DV, Pandit AB, Shankarling GS (2011) Synthesis of chalcone (3-(4-fluorophenyl)-1-(4-methoxyphenyl) prop-2-en-1-one): Advantage of sonochemical method over conventional method. Ultrason Sonochem. https://doi.org/10.1016/j.ultsonch.2010.09.010
46. Figueroa-Ariza LT, Paez-Sierra BA (2026) Solvent-Dependent Spectral Deconvolution of Amino-Substituted Chalcones: UV–vis and FT-IR Analysis Supported by TD-DFT Calculations. J Mol Struct. https://doi.org/10.1016/j.molstruc.2025.144083
47. Turczyn R, Krukiewicz K, Katunin A, Sroka J, Sul P (2020) Fabrication and application of electrically conducting composites for electromagnetic interference shielding of remotely piloted aircraft systems. Compos Struct. https://doi.org/10.1016/j.compstruct.2019.111498
48. Shirani M, Akbari-adergani B, Rashidi Nodeh H, Shahabuddin S (2020) Ultrasonication-facilitated synthesis of functionalized graphene oxide for ultrasound-assisted magnetic dispersive solid-phase extraction of amoxicillin, ampicillin, and penicillin G. Microchim Acta. https://doi.org/10.1007/s00604-020-04605-z
49. Passador FR, de Faria PV, Backes EH, Montanheiro TL, Montagna LS, de Souza Pinto S, Pessan LA, Rezende MC (2017) Thermal, mechanical and electromagnetic properties of LLDPE/PANI composites. Polym Bull. https://doi.org/10.1007/s00289-016-1862-5
50. Fang Q, Chen B, Zhuang S (2013) Triplex Blue-shifting Hydrogen Bonds of ClO_4^- ···H–C in the Nanointerlayer of Montmorillonite Complexed with Cetyltrimethylammonium Cation from Hydrophilic to Hydrophobic Properties. Environ Sci Technol. https://doi.org/10.1021/es402490k
51. Barnes AJ (2004) Blue-shifting hydrogen bonds—are they improper or proper? J Mol Struct. https://doi.org/10.1016/j.molstruc.2004.02.040
52. Hansen PE, Spanget-Larsen J (2017) NMR and IR Investigations of Strong Intramolecular Hydrogen Bonds. Molecules. https://doi.org/10.3390/molecules22040552
53. Joseph J, Jemmis ED (2007) Red-, Blue-, or No-Shift in Hydrogen Bonds: A Unified Explanation. J Am Chem Soc. https://doi.org/10.1021/ja067545z
54. Stejskal J, Trchová M, Bober P, Humpolíček P, Kašpárková V, Sapurina I, Shishov MA, Varga M (2015) In: Mark HF (ed) Encyclopedia of Polymer Science and Technology. Wiley, New York. https://doi.org/10.1002/0471440264.pst640.
55. Shayegh S, Amrollahi Bioki H, Zarandi MB, Samani NK, Rahnamanic A (2017) ZnS nanoparticles incorporated in polyaniline composite: preparation and optical characterization. Polym Sci Ser B. https://doi.org/10.1134/S1560090417050104
56. Patil PT, Anwane RS, Kondawar SB (2015) Development of electrospun polyaniline/ZnO composite nanofibers for LPG sensing. Procedia Mater Sci. https://doi.org/10.1016/j.mspro.2015.06.041
57. Chatterjee S, Shit A, Nandi AK (2013) Nanochannel morphology of polypyrrole–ZnO nanocomposites towards dye sensitized solar cell application. J Mater Chem A. https://doi.org/10.1039/c3ta12796f
58. Myint MTZ, Hada M, Inoue H, Marui T, Nishikawa T, Nishina Y, Ichimura S, Umeno M, Kyaw AKK, Hayashi Y (2018) Simultaneous improvement in electrical conductivity and Seebeck coefficient of PEDOT: PSS by N_2 pressure-induced nitric acid treatment. RSC Adv. https://doi.org/10.1039/C8RA06094K
59. Al-Asbahi BA, El-Shamy AG (2023) The thermoelectric power function of new composite Poly (3, 4-ethylenedioxythiophene): poly (styrene sulfonate)/carbonyl iron films: A new exploration. Synth Met. https://doi.org/10.1016/j.synthmet.2023.117280
60. Fan Z, Ouyang J (2019) Thermoelectric properties of PEDOT: PSS. Adv Elect Mater. https://doi.org/10.1002/aelm.201800769
61. Arslanturk G, Yildirim S, Sen E, Karadeniz SC, Ugraskan V, Isik B, Erdogan Cakar A, Cakar F, Cankurtaran H, Cankurtaran O, Yazici O (2026) Nickel sulfide-filled polyaniline hybrid composites for high-performance thermoelectric materials at room temperature. Synth Met. https://doi.org/10.1016/j.synthmet.2025.118048

62. Lin Y, Shi Z, Wildfong PLD (2010) Thermal conductivity measurements for small molecule organic solid materials using modulated differential scanning calorimetry (MDSC) and data corrections for sample porosity. J Pharm Biomed Anal. https://doi.org/10.1016/j.jpba.2009.10.022
63. Pokharel M, Zhao H, Ren Z, Opeil C (2013) Grain boundary Kapitza resistance analysis of nanostructured $FeSb_2$. Int J Therm Sci. https://doi.org/10.1016/j.ijthermalsci.2013.03.009